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RV COLLEGE OF ENGINEERING
Autonomous Institution affiliated to VTU
IV Semester B.E. September-2024 Examinations
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
SIGNALS AND SYSTEMS
(2022 SCHEME)

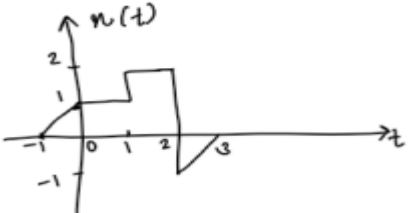
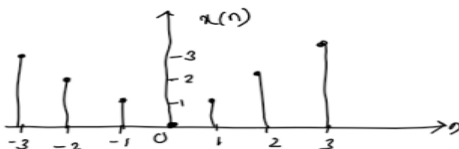
Time: 03 Hours**Maximum Marks: 100****Instructions to candidates:**

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, and 9 and 10.

PART-A

1	1.1	The fundamental period of the signal $8\sin(0.8\pi t + \pi/4)$ is _____ .	1
	1.2	If the Energy of a signal $x(t)$ is E, Then Energy of the signal $x(2t)$ is _____.	1
	1.3	Determine the system $y(t) = \cos x(t)$ is time variant or time invariant.	1
	1.4	The FT of $3\text{sinc}3t$ has amplitude _____ and period _____ .	1
	1.5	What is the speed improvement factor in terms of complex multiplications for the direct computation of the DFT over FFT approach for $N = 64$?	1
	1.6	A complex valued sequence $x[n]$ of length $N=256$ is to be convolved with a complex valued sequence $h[n]$ of length $L=64$. Find the number of complex multiplications required to perform this convolution directly.	1
	1.7	Consider the following data set $x = [1, 1, 1, 1, 1, 1, 1, 1]$, what is the value of skewness and its distribution?	2
	1.8	Evaluate the step response of the system if impulse response is given by $h(n) = \delta(n) - \delta(n-1)$.	2
	1.9	A 5-point sequence $x(n)$ is given as $x[-3]=1$, $x[-2]=1$, $x[-1]=0$, $x[0]=5$ and $x[1]=1$. If $X(e^{j\Omega})$ denotes the discrete-time Fourier transform of $x(n)$. Then the value of is _____.	2
	1.10	Consider $x(n) = (1/2)^n u(n)$, $y(n) = x^2(n)$, and $Y(e^{j\Omega})$ is the Fourier transform of $y(n)$, then $Y(e^{j0}) =$ _____ .	2
	1.11	Consider a sequence $x[n] = \{1, 2, 0, 1\}$. Find the DFT of the sequences $x(n)$ and $y(n) = x((-n))_4$.	2
	1.12	The circular autocorrelation of the sequence $r[n] = \{1, 0, 1, 0\}$ is _____.	2
	1.13	If 8-point DFT of a sequence $p(n)$ is $P(k) = \{8, 1 + j1, 2 + j2, 1 + j2, 4, 1 - j2, 2 - j2, 1 - j1\}$ then the function $\sum_{n=0}^7 e^{j\pi n} p(n)$ is _____.	2

PART-B

UNIT-I			
2	a	Find energy and power of the following signals: i. $x(t) = u(t)$ ii. $x(t) = 4 e^{j9t}$	4
	b	Consider the signal shown in Fig. 2b, Sketch $y(t) = [x(t) + x(2-t)] - u(1-t)$.  Fig. 2b	6
	c	Consider the signal shown in Fig. 2c.  Fig. 2c Sketch (i) $x(n)\{u(n-3) - u(n-1)\}$ (ii) $x(n)\delta(n-1)$ (iii) $x(2-n)u(n-1)$	6
UNIT-II			
3	a	Evaluate the convolution integral of $f(t)$ and $g(t)$ shown in fig. 3a  Fig. 3a	8
	b	The output of a discrete-time system is related to its input $x[n]$ as follows: $y[n] = 2 x[n] + 4 x[n-2] + 6 x[n-4] + 8x[n-6]$ i. Formulate the operator for the system-relating $y[n]$ to $x[n]$. ii. Develop the block diagram representation for H , in both cascade and parallel implementation.	4
	c	Determine whether the given impulse response of CT system $h(t) = e^{-5 t }$ is memory less, causal and stable.	4

OR			
4	a	Investigate the causality and stability of the following LTI system: (i) $h(n)=2^n u(n-1)$ (ii) $h(n)=(0.5)^{ n }$	4
	b	Find the step response of an LTI system, whose impulse response is given by $h(t)=e^{- t }$.	5
	c	Compute the convolution sum of $x(n)=(0.5)^n u(n)$ and $h(n)=\delta(n)-0.5\delta(n-1)$.	7

UNIT-III			
5	a	Let the input to a discrete LTI system be $x(n) = 10 + \cos(\pi n + \pi/3)$. Determine the output of this system for the following impulse response, $h(n) = (-1)^n \frac{\sin(\frac{\pi}{6}n)}{\pi n}$. Also, sketch $X(e^{j\Omega})$, $H(e^{j\Omega})$ and $Y(e^{j\Omega})$.	10
	b	Show that the Fourier transform of Dirac comb is also a Dirac comb.	6
OR			
6	a	Let the input to a discrete LTI system be $x[n] = \cos(\pi n/8) + \sin(3\pi n/4)$. Determine the output of this system for the impulse response $h[n] = (-1)^n \sin(\pi n/2)/\pi n$.	8
	b	Find the DTFS and DTFT representation of the periodic signal. Sketch magnitude and phase spectrum.	8

UNIT-IV			
7	a	Consider the sequences $x_1(n)=\{0,1,2,3,4\}$ a. Sketch the sequence $s(n)$ with six-point DFT if $S(K) = W_2^k X(K)$ b. Determine the sequence $y(n)$ with six-point DFT $Y(K)=R(X(K))$ c. Determine the sequence $v(n)$ with six-point DFT $V(K)=I(X(K))$	8
	b	Consider $x(n) = \{4, 8, 8, -6, 8, 3, 4, -3, 6, 7, 3, 0.5, -1.5, 1, 9, 1, 3, 5\}$. What is the response of the system with impulse response $h(n) = \{-7, 9, 5\}$ using the overlap add method? Assume block length of 6.	8
OR			
8	a	Find 4-point circular convolution of $x(n) = \{4, 3, -2, -1\}$ and $h(n) = \{-1, -2, 3, 3\}$ using DIF-FFT algorithm.	8
	b	Let $p(n) = \{3, 1, 0, 1, 3, 2, 0, 1, 2, 1, 0, 1\}$, be a long sequence obtained by sampling a continuous time domain signal. The signal is passed through an LTI system with impulse response $h(n) = \{1, 1, 1\}$. Find the output of the system using overlap-add method with 5-point circular convolution.	8

UNIT-V			
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9	a	Describe the concept of skewness and kurtosis in time domain analysis. How can positive and negative skewness values indicate different characteristics of a signal and relate to the distribution of data points with an example.	8
	b	Write a short note on classification of signals based on time domain feature extraction process with relevant examples.	8
OR			
10	a	Write a short note on classification of signals based on frequency domain feature extraction process with relevant examples.	8
	b	Consider the dataset representing signal values over time and the values are [22, 25, 20, 12, 15, 30, 200, 18, 15, 100, 28, 23, and 24], find the outliers in the given data set using IQR method.	8